

Blockchain Technology — Course Syllabus

Program	Bachelor of Computer Applications	Semester	V
Year	III	Course Name	Blockchain Technology
Code	BCADSN25326	Course Type	DSE
Pre-Requisite	2	L	2
		T	1
		P / Credit	0 / 3

Course Objectives: To gain a comprehensive understanding of Blockchain and Distributed Ledger Technologies, covering fundamental concepts and functionalities. Delve into Alternative Blockchain to grasp the workings of Distributed Ledger Technology beyond conventional paradigms.

Course Outcomes

CO1	Students will learn fundamental concepts of Blockchain and Distributed Ledger Technologies.
CO2	To acquire the insights into Blockchain functionality.
CO3	To explore Blockchain implementation through Bitcoin and Merkle Root etc.
CO4	To get knowledge about Distributed Ledger Technology in Alternative Blockchains.

Module	Course Contents	Contact Hrs.	Mapped CO
1	Blockchain and Distributed Ledger Fundamentals: Blockchain, Growth of Blockchain technology, Cryptographic basics for cryptocurrency: signature schemes, encryption schemes; Categories of Blockchain: Public Blockchain, Private Blockchain, Permissioned Ledger, Tokenized Blockchain, Tokenless Blockchain.	10	CO1
2	Blockchain Functionality: Distributed identity and Digital identification: Public and private keys, Decentralized network, Permissioned distributed Ledger, Digital identification and wallets; Blockchain data structure and security: Double spending, Network consensus, Sybil attacks, Block rewards and miners, Forks and consensus chain, Sharding based consensus algorithms to prevent attack, Finality, Limitation of proof-of-work, Alternatives to Proof of Work.	12	CO2
3	Blockchain Implementation: Bitcoin and Merkle Root; Eventual Consistency and Bitcoin; Byzantine Fault Tolerance and Bitcoin; Bitcoin block-size; Bitcoin Mining; Blockchain Collaborative Implementations: Hyperledger, Corda; Ethereum's ERC-20 and token explosion; Blockchain and full ecosystem decentralization: Smart contract, Decentralized autonomous organization (DAO), Decentralized applications.	12	CO3
4	Distributed Ledger Technology in Alternative Blockchain: Blockchain Governance Challenges: Bitcoin Blocksize Debate, The Ethereum DAO Fork, Ethereum's Move to PoS and Scaling Challenges; Blockchain Technical Challenges: Denial-of-Service Attacks, Security in Smart Contracts, Ripple, Stellar; Decentralized Network manager: Tezos.	11	CO4